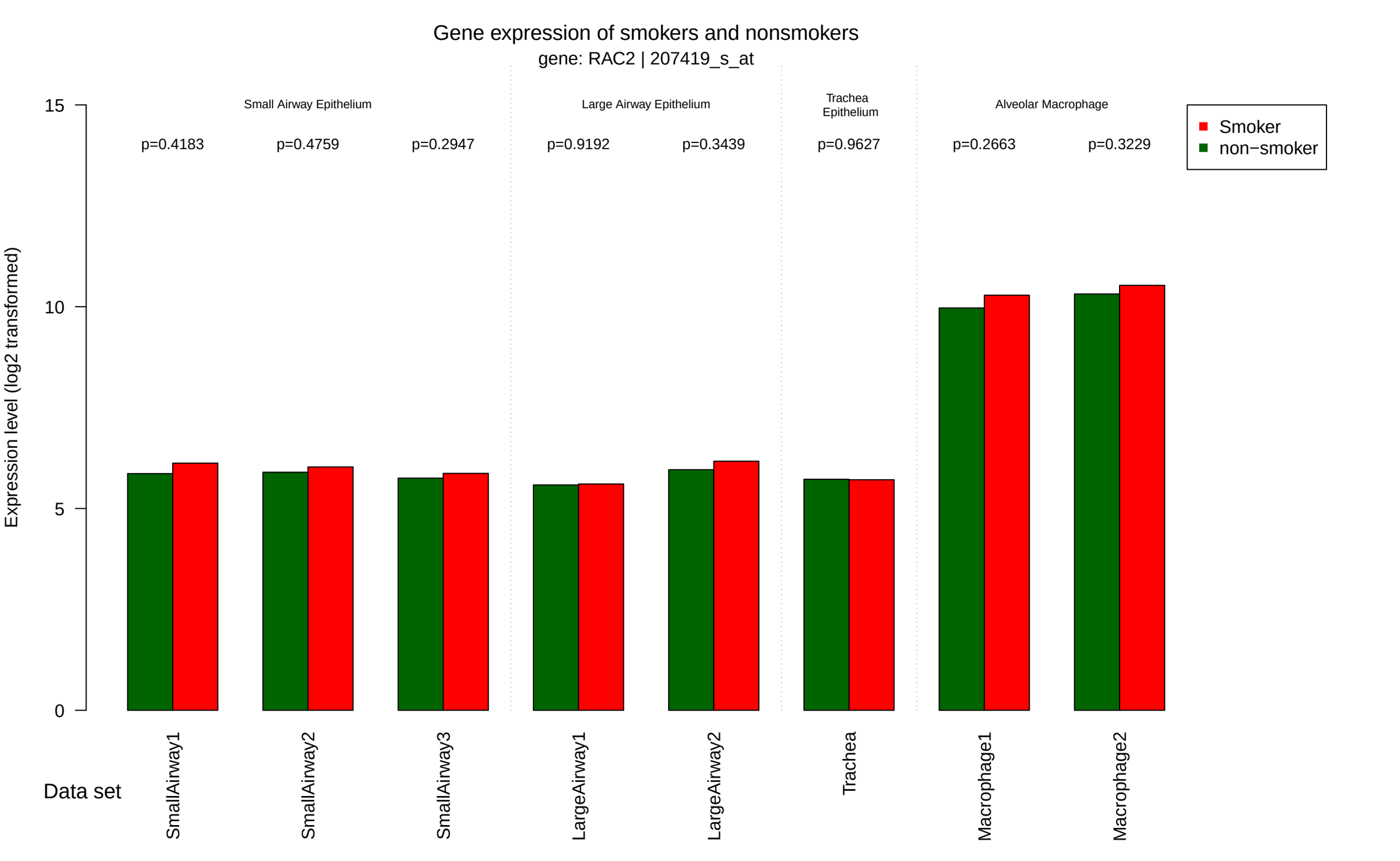
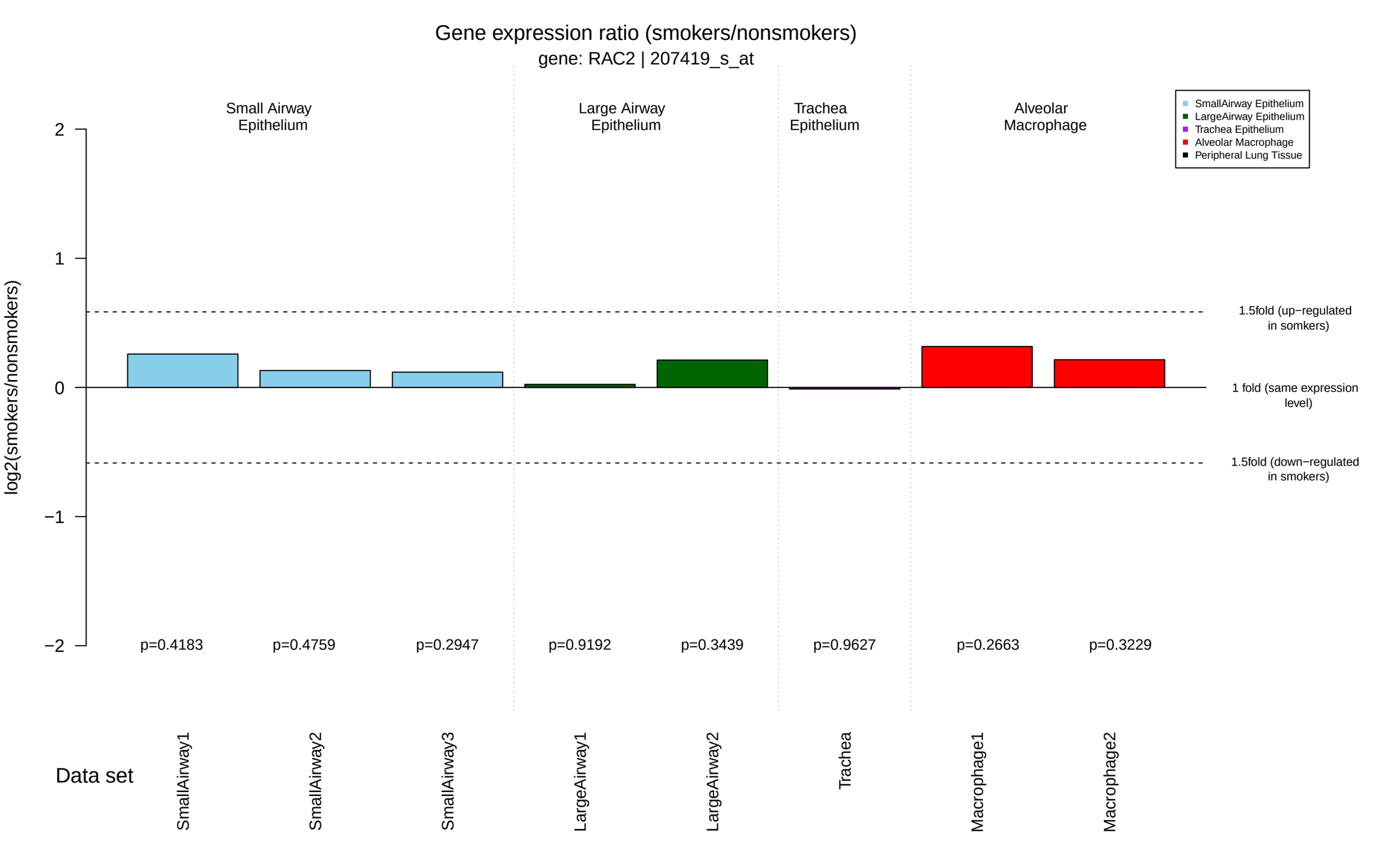
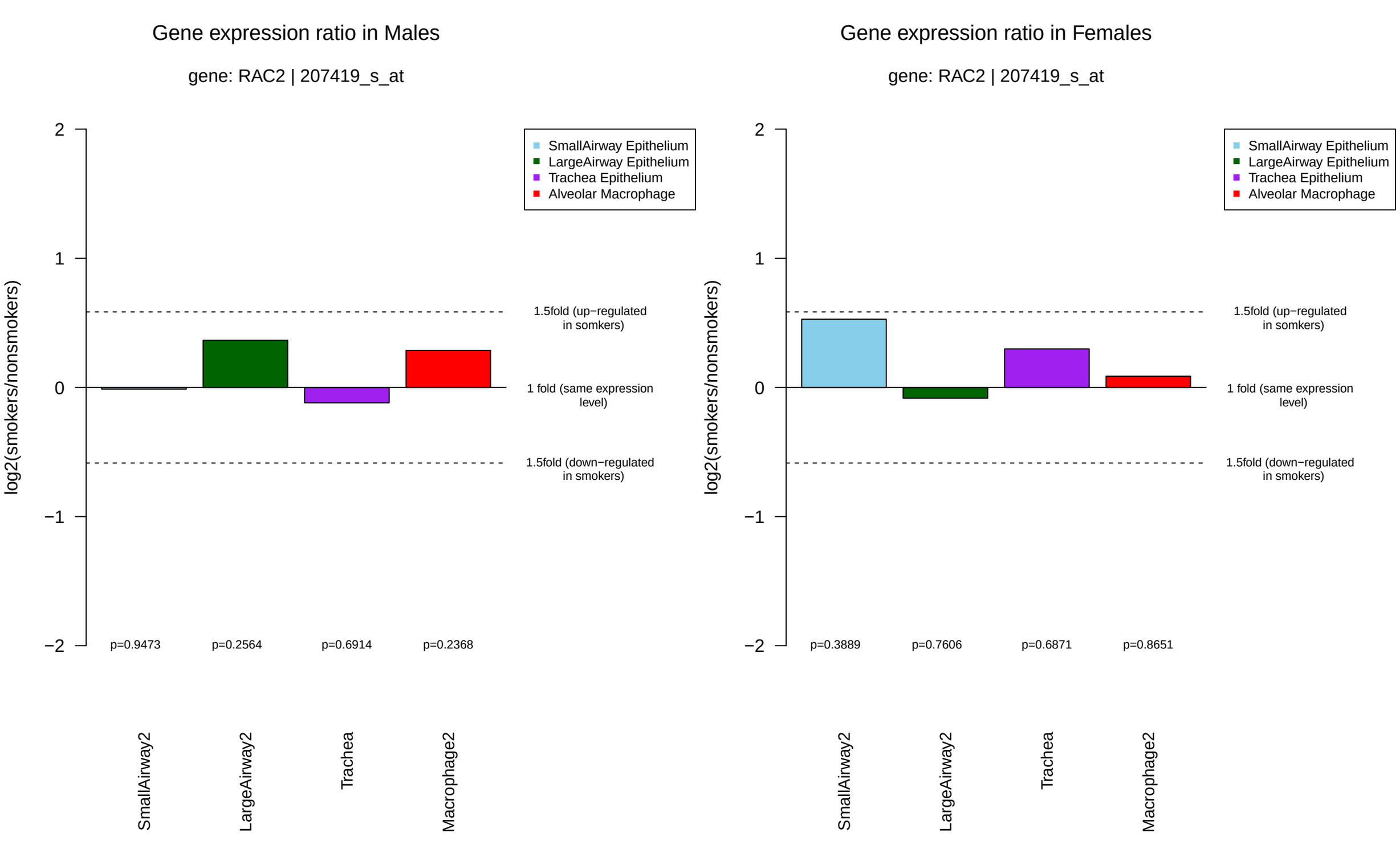
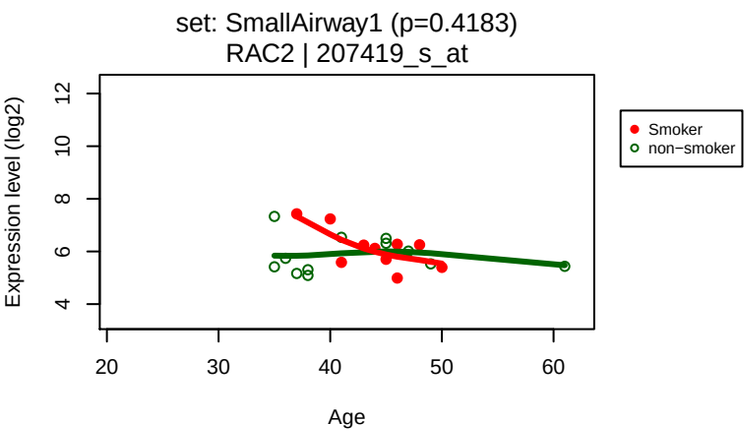
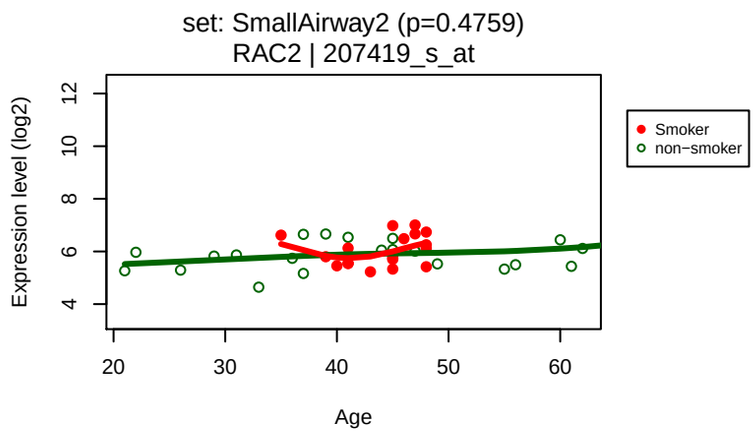
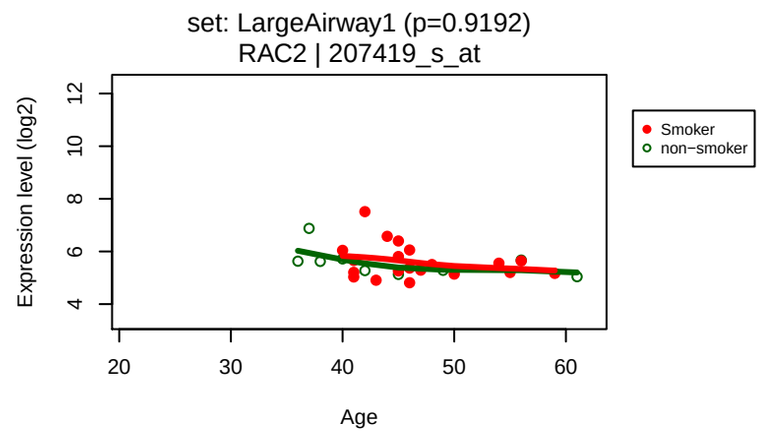

Gene expression of smokers and nonsmokers
gene: RAC2 | 207419_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4183
p=0.4759
p=0.2947
p=0.9192
p=0.3439
p=0.9627
p=0.2663
p=0.3229
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

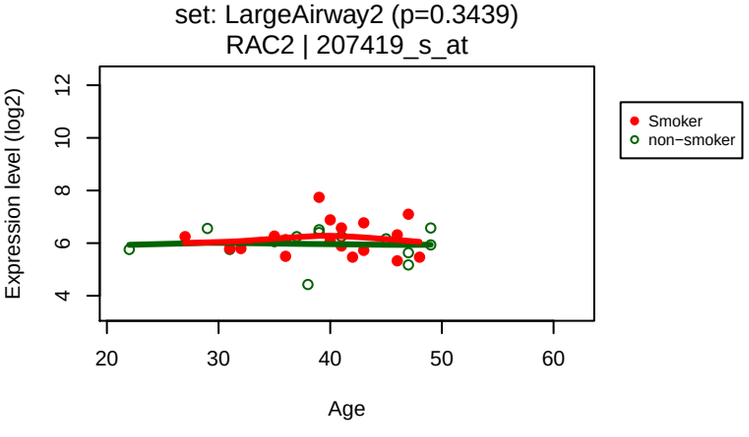

Gene expression ratio (smokers/nonsmokers)
gene: RAC2 | 207419_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4183
p=0.4759
p=0.2947
p=0.9192
p=0.3439
p=0.9627
p=0.2663
p=0.3229
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

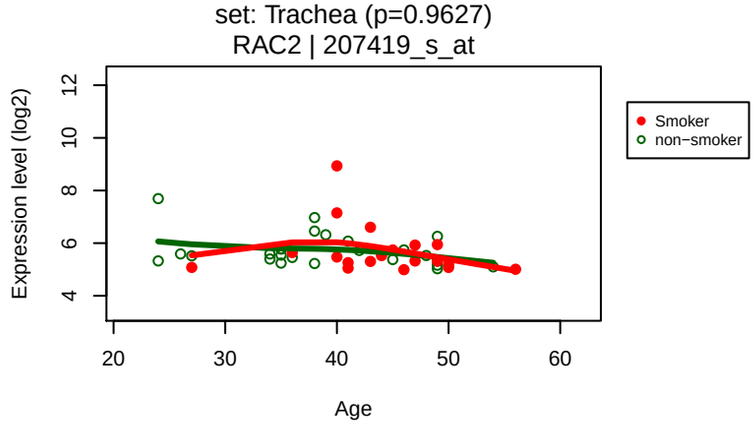

Gene expression ratio in Males
gene: RAC2 | 207419_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9473
p=0.2564
p=0.6914
p=0.2368
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAC2 | 207419_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3889
p=0.7606
p=0.6871
p=0.8651
SmallAirway2
LargeAirway2
Trachea
Macrophage2

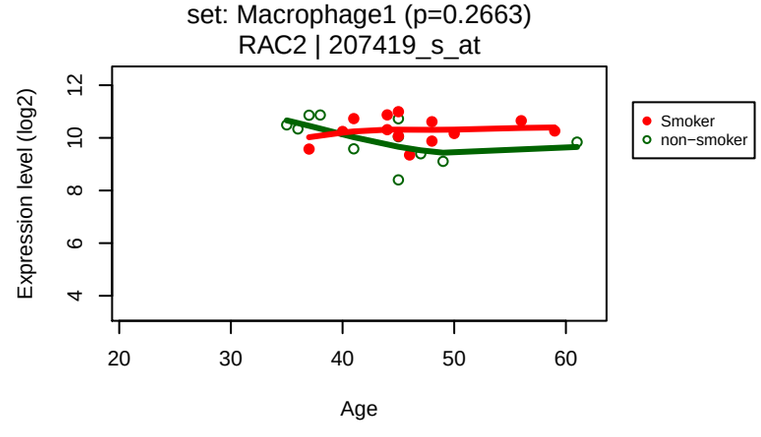

set: SmallAirway1 (p=0.4183)
RAC2 | 207419_s_at
Expression level (log2)
Age
Smoker
non−smoker

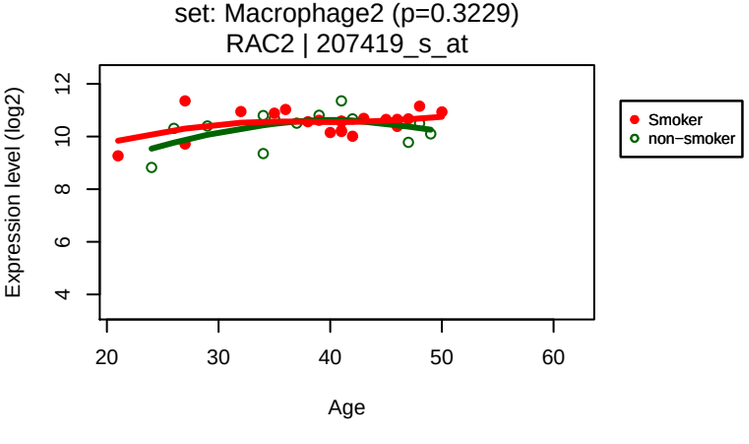

set: SmallAirway2 (p=0.4759)
RAC2 | 207419_s_at
Expression level (log2)
Age
Smoker
non−smoker

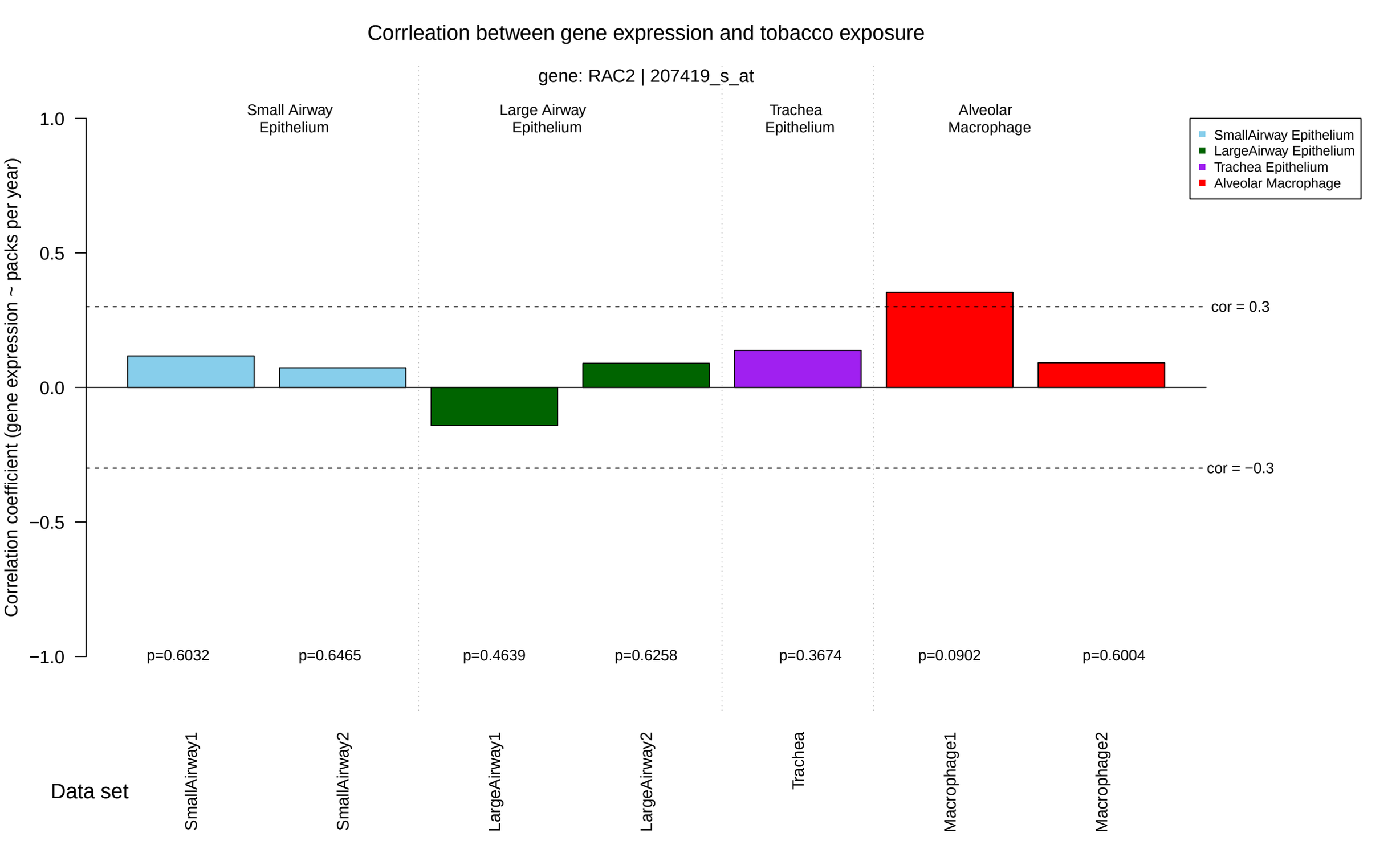

set: LargeAirway1 (p=0.9192)
RAC2 | 207419_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3439)
RAC2 | 207419_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9627)
RAC2 | 207419_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2663)
RAC2 | 207419_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3229)
RAC2 | 207419_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAC2 | 207419_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6032
p=0.6465
p=0.4639
p=0.6258
p=0.3674
p=0.0902
p=0.6004
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2